Eliciting the emotion PLAY in Virtual Reality

Literature Review

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Abstract

This literature review examines the potential of Virtual Reality (VR) as a medium for emotion elicitation, with particular focus on Panksepp's primary emotional system of PLAY and its therapeutic implications for depression. Existing literature on VR-based positive emotion induction is critically reviewed to establish the theoretical and empirical foundations for this research. A significant gap is identified: while VR has been demonstrated to elicit a range of positive emotional states, no study has specifically designed a VR environment to target Panksepp's PLAY system as a discrete primary emotion. This review proposes that such an investigation is both theoretically motivated and clinically relevant, and outlines a methodological framework combining self-report measures and physiological data collection through which this gap may be addressed.

Keywords

Depression, Virtual Reality, PLAY, Virtual Environments

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1 Introduction and Motivation

Virtual Reality (VR) is a tool that has been in development for over six decades. Its modern application as an emotion-elicitation tool relies on its unique ability to foster 'Presence' [13]. This is the psychological state of being 'there' within the digital environment. Unlike other classical methods, VR allows users to be enveloped in the experience and allows them to have some degree of freedom when interacting with the Virtual Environment (VE) [8]. Central to this literature review is Jaak Panksepp's PLAY system, an emotion associated with rough-and-tumble play such as the one children take part in when they play hide-and-seek [19]. Given that the depressive affect is associated with an overactive GRIEF system, PLAY offers a promising, yet under-researched, therapeutic path [19]. This literature review aims to evaluate the methodologies for measuring positive affective states, the effectiveness of VR as an emotion elicitation medium and to determine whether PLAY would be effective in the treatment of depression. It will also discuss the weaknesses and limitations of similar work.

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2 PLAY

2.1 Jaak Panksepp's Emotional Frameworks

Jaak Panksepp defined seven emotions in his affective neuroscience emotional framework. There are four positive emotions namely, PLAY, CARE, LUST, SEEKING and three negative emotions namely, RAGE, FEAR and GRIEF [20].

SEEKING traditionally called the "brain reward system" is a general-purpose appetitive motivational system that is essential for animals to acquire all resource needs for survival. It promotes learning by mediating anticipatory eagerness, partly by coding predictive relationships between events. It becomes underactive with accompanying feelings of depression while overactivity promotes excessive and impulsive behaviors [20].

RAGE is an emotion that is provoked when an animal is prevented from doing what it wants to do. It lies close to and interacts with trans-diencephalic FEAR systems highlighting the source of the term "fight or flight" [20].

FEAR is an important emotion for protecting animals against pain and destruction [20].

LUST is an emotion that is aroused by male and female sex hormones which help create gender specific tendencies. In females, the sex hormone estrogen promotes trust and confidence while in males, the sex hormone vasopressin promotes assertiveness. The dopamine-driven SEEKING participates in the search of sexual rewards [20].

CARE is the affiliative emotional system that motivates nurturing, protective behaviours toward offspring and other vulnerable individuals [20].

GRIEF on the other hand, initially known as the PANIC system is an emotion that is triggered when secure attachments are broken such as when a young offspring is lost [20].

Lastly the emotion PLAY is characterised by strong urges for physical play such as running, chasing, pouncing and wrestling. These actions are accompanied by positive affect, an intense social joy [20]. Each of the seven primary emotional systems can be distinguished by its underlying neuroanatomy, the conditions under which it is provoked, and its evolutionary function. These distinctions are summarised in Table 1.

2.2 What is PLAY

PLAY is one of the affective emotions as described by Jaak Panksepp. It is also referred to as social joy and capitalised to distinguish it from the common term play [4]. PLAY is the emotion that stimulates especially young animals to engage in physical activities like wrestling, running, and chasing each other which helps in

Table 1: Summary of emotions, areas responsible for the emotion, how it is provoked and the importance of the emotion

Emotion	Neuroanatomical substrate	Provocation conditions	Evolutionary function
SEEKING	Medial forebrain bundle	Anticipation, novelty and the prospect of a reward	Coaxes animals to acquire everything they need for survival
RAGE	Medial amygdala and hypothalamus	Curtailing animals' freedom of action	Helps animals defend themselves
FEAR	Central amygdala, the anterior and medial hypothalamus	Unconditional threats	Protection from pain and destruction
LUST	Preoptic area of the hypothalamus [4]	Male and female sex hormones	To drive reproduction and ensure survival of species [4]
CARE	The hypothalamus, the Anterior Cingulate Cortex (ACC), the amygdala and basal forebrain	Shifting hormonal tides at the end of pregnancy	Ensures that parents take care of their offspring
GRIEF	Dorsal PAG to anterior cingulate	The major-social-attachment, social bonding chemistries of the mammalian brain such as oxytocin	Foundational for the secure attachments that are essential for future mental health and happiness
PLAY	Subcortical circuits within the periaqueductal gray area (PAG) and the lower limbic system	Rough and tumble play such as running, chasing, pouncing	Learn social rules and refine social interactions

social bonding [4]. It is considered as one of the positive emotions alongside LUST, CARE and SEEKING.

2.3 The Importance of PLAY

Depressive affect is caused by overactivity of GRIEF leading to psychological pain and underactivity of SEEKING causing a feeling of giving up [20]. Therapeutic approaches that promote the positive hedonics of social CARE and PLAY systems may provide better treatment options resulting in better outcomes [20]. Given that PLAY can promote the expression of various neurotrophins such as brain-derived neurotrophic factor, it is expected that playful interactions have anti-depressant effects similar to exercise [20]. Research also suggests that certain discrete positive emotions such as PLAY have the ability to broaden people's momentary thought-action repertoires and build their enduring personal resources [9]. The broadening of intellectual resources increases levels of creativity, promotes creation of theory of mind, and fuels brain development [9]. For our ancestors, broadening of physical resources as a result of PLAY increased chances of survival and in turn greater odds of living long enough to reproduce [9]. PLAY also facilitates the maturation of the frontal lobe inhibitory skills that gradually come to regulate children's impulsive primary-process emotional urges [18]. Playful activities decrease amygdala activation which prevents the frontal lobe from being overpowered leading to emotional reactivity rather than thoughtful action [3]. PLAY offers a promising alternative treatment option for depression due to the underlying neuroanatomy.

2.4 Measuring PLAY

The elicitation of PLAY can be measured through both physiological markers and behavioral observations. In humans, positive affect can be inferred through the Duchenne smile, as research shows a strong correlation between this expression and self-reported positive affect on Likert-type scales [3]. Neurally, the level of positive stimulation can be determined by measuring amygdala activation, as decreased activation in this region correlates with positive emotional states [3].

Beyond human studies, Panksepp et al. identified behavioral markers of PLAY in animal models. Observing the rough-and-tumble play of adolescent rats, researchers discovered that these interactions evoked 50-kHz "chirpy" laughter-like vocalizations [3]. These calls reflect an appetitive positive affective state similar to primitive human joy and laughter [3].

In addition to these biological markers, PLAY is assessed using two primary self-report scales. The first is the Affective Neuroscience Personality Scale (ANPS), which focuses on the ancient brain processes that serve as a foundation for "higher" mental attributes [5]. The second is the Five Factor Model (FFM), which has been used to validate the ANPS [5]. Items on the ANPS are designed to access personal feelings and behaviors rather than cognitive social judgments, using statements such as, "I am known as the one who keeps work fun" [5]. In the study by Davis et al., the scale was limited to fourteen items, seven positive and seven reversed-scored, with filler items interspersed for validity checks [5].

The standard order of the ANPS scale is SEEK, FEAR, CARE, ANGER, PLAY, and SADNESS. Following data collection from both the FFM and ANPS, reliabilities are typically computed using Cronbach's alpha. Exploratory analysis has revealed significant relationships between these scales; specifically, extraversion on the FFM correlates strongly with the PLAY system [15]. Montag et al. argue that these primary emotions form the foundation of personality and serve as a template for expanding our interpretation of the Big Five [15]. However, a key limitation of the current ANPS is its focus on stable emotional traits rather than momentary states [15]. Consequently, there is a documented need for a state-version of

the ANPS to capture emotions as they occur in response to specific stimuli [15].

2.5 Joy versus PLAY

The "dimensional" approach represents emotional states as locations within a continuous affective space. This space is typically organised along two primary axes: arousal, capturing how activated or energised an individual feels, and valence, capturing whether the experience is positive or negative [20]. Under this framework, joy and PLAY would not be considered distinct emotions but rather neighbouring points along these shared dimensions. This stands in contrast to Jaak Panksepp's hierarchical model of affective processing, which proposes three neuroanatomically distinct levels of emotional organisation: the primary level, engaging subcortical networks, the secondary level, engaging the limbic system, and the tertiary level, engaging the neocortex [17]. Within this hierarchy, PLAY is characterised as a primary process, while joy also referred to as happiness operates at the secondary and tertiary levels [2]. Neuroanatomically, joy engages parts of the limbic system and the frontal neocortical structures, whereas PLAY engages emotion-specific subcortical circuits within the periaqueductal gray area (PAG) and the lower limbic system [2]. This hierarchical distinction has direct implications for measurement, as instruments used to measure secondary or tertiary level emotions may not adequately capture primary-level emotions such as PLAY.

PLAY is measured using the Affective Neuroscience Personality Scale (ANPS), a self-report instrument specifically designed to assess individual differences in Panksepp's six primary emotional systems [5]. Joy, by contrast, is most commonly measured using one of two questionnaires, each with distinct strengths and limitations as summarised in Table 2.

3 Emotion Elicitation

Emotions play a significant role in human experience; as a result, scientists have refined methods for eliciting, assessing, and analysing emotions in order to improve their understanding[23]. Eliciting emotions in a controlled setting is valuable to researchers as they are able to isolate specific affective states. Furthermore, they can study how those affective states affect the behavior of human beings such as the impact on decision making skills. This is why using the right elicitation methods ensures the success and quality of research, and to draw reliable conclusions [27]. Emotions can be elicited actively or passively. Passive emotion elicitation involves the use of media such as videos, films and images to elicit the target emotion. Although the use of video clips has reported a one hundred percent accuracy in predicting target emotions, passive elicitation limits the generation of reliable theoretical frameworks [7]. This is due to the fact that the role of the subject is an observer who does not have any involvement on what is happening. Active emotion elicitation mediums such as the use of VR and the interactions with humans and computer-generated avatars that respond with facial expressions and behavioral changes, allow the subject to have a more immersive experience and consequently results in the generation of more reliable theoretical frameworks [27]. Although film and story-based protocols are established as the most effective passive elicitation techniques [28], Virtual Reality provides a superior active elicitation framework for emotional engagement. The interactive nature of VR grants the user narrative agency, a critical component for eliciting complex primary emotions like PLAY that require active participation rather than mere observation.

3.1 Ethical considerations in Emotion elicitation

When obtaining biometric data, such as an electroencephalogram (EEG), to study emotion elicitation, ethical concerns regarding data collection and storage must be addressed to safeguard participant autonomy and privacy. The implementation of ethics-focused checklists can guarantee that all data is handled according to established standards [1]. It is equally important to minimize bias by representing diverse groups, which validates that the findings apply to a broader population [1]. Furthermore, explaining the collection process including usage, storage, and sharing procedures while allowing participants to withdraw without consequence maintains their autonomy [1]. To address psychological implications, providing debriefing sessions and mental health support resources is vital [1]. Before beginning the study, researchers must clearly explain potential risks to allow participants to decide if taking part is in their best interest, a more ethical approach than simply excluding individuals based on rigid criteria [12].

4 Virtual Reality as a Positive Emotion Inducer

4.1 Immersion, Agency and Presence

There are three important aspects of Virtual Reality Experiences. The first is 'Immersion'. Immersion is described as what the technology delivers (in all sensory modalities) from an objective point of view. More sophisticated technology such as the use of high-quality head mounted displays (HMD) result in higher levels of immersion [6]. Research also suggests that strongly arousing emotions, such as fear are stronger in immersive setups compared to weaker arousing emotions such as happiness in the same setup [6].

The second aspect of VR experiences is Presence. Presence refers to the illusion of being in a different environment [24]. Ensuring that a VR experience creates a high level of Presence is paramount due to the correlation between the level of Presence and the emotional experience. More specifically, high levels of immersion result in higher levels of presence in a VR experience, which in turn affects the emotional experience [6]. Although the level of immersion in a VR exerts an effect on presence, some research suggests that this is strongest when there is no emotion involved [6].

The third important aspect of VR experiences is Agency. Agency is the satisfying power the user has to take meaningful actions and see the results of their decisions and choices. The more realised the immersive environment, the more agency a user would want [16]. Agency allows the user to feel like they own the narrative in the VR, which ultimately affects their emotional experience.

4.2 How Effective is VR in Eliciting Emotion?

In contrast to classical methods such as the use of film, Virtual Reality (VR) serves as a potent medium for emotion elicitation by simultaneously engaging the psychological and physiological components of the subject [25]. The efficacy stems from a high

Table 2: Comparison of questionnaires used to measure joy

Instrument	Construct measured	Strengths	Limitations
Positive and Negative Affect Schedule (PANAS)	Used to measure responses to situational manipulations of emotion [11]	It is well established and widely validated [11]	Assesses broad categories of positive and negative affect [11]
Discrete Emotions Questionnaire (DEQ)	Used to measure discrete emotions [11]	Validated to reliably detect eight discrete emotional states: anger, disgust, fear, anxiety, sadness, happiness, relaxation, and desire [11]	While the DEQ does not include a dedicated PLAY subscale, it measures joy and happiness as the closest available approximation to Panksepp's PLAY system [11]

degree of 'Presence' during a VR experiment which has been found to be an important evaluation indicator in the VR experience as a mental state [13].

4.3 How is VR Used in Emotion Elicitation?

Dozio et al. suggest that designing a Virtual Environment (VE) for emotion elicitation is not dissimilar to designing a complex and multisensory interactive real experience. The designs of VEs aimed at emotion invocation require that different aspects such as the user experience, sensory channels and content are combined to create a high degree of presence, immersion and agency. All of which are important for the VE to be successful. The use of design elements associated with a specific emotion leads to a successful Virtual Environment. For example, if one is trying to elicit fear, a shark element would be a crucial element to consider. Interaction in VRs is also an important element as it allows the user to control the narrative and as a consequence feel like they own the narrative. In a VE, the environment should have influence on the user and the user should have influence on the environment [8]. For effective emotion elicitation in Virtual Environment, there are four important elements to consider. The first is the semantic elements which aim to answer the question 'What should the environment communicate to users so that they feel a specific emotion?'. The use of symbolic elements or elements based on real-life experiences leads to high arousal. The second is the sensory elements divided into visual, sound and touch. Color, Lighting, Textures and shapes are examples of things considered when designing sensory elements. The third is dynamics. This refers to how the objects in the VE move relative to the user such as the speed of the user or the way the user jumps. It is vital in delivering an experience that is full of events. The last element is interactive elements. Giving the user guidance and exploration cues invites the users to perform actions and movements which supports agency. Careful consideration is made when designing these four elements to lead to high arousal [8].

Based on the analysis of literature where VR was successfully used to elicit emotion, VR based media content can be divided into four main categories including visual or audio-visual stimuli, games and tasks, 360-degree media and mixed reality [27]. Careful consideration is made to ensure participants' are exposed to the same stimuli such as screen brightness and screen size in order to eliminate any biases. The use of virtual nature as part of the content in the virtual environment has been confirmed to increase positive emotions [21]. The use of personalised 360 degrees videos has also been found to be effective in inducing positive emotions. Furthermore, studies have found that the use of auditory stimuli mostly consisting of music and ambient sounds such as bird chirping results in increased positive emotions [21].

4.4 Measuring PLAY and Other Emotions Elicited by VR

Given that no study to date has specifically investigated the use of VR to elicit Panksepp's PLAY system, the following section examines the broader literature on VR-based emotion elicitation, with particular attention to the methodological approaches and measurement instruments employed in these studies.

Researchers measure emotions elicited by VR in two ways, by obtaining subjective data and obtaining objective data. Subjective data is obtained through the use of questionnaires given to the participants such as the 9-point SAM scale which is widely used to measure the valence, arousal and dominance associated with a person's affective reaction. The Affective Neuroscience Personality Scales (ANPS) is also a self-report tool that assesses individual differences in Panksepp's primary emotional systems [14]. While the PANAS-X scale is designed to assess 11 specific emotional states of Fear, Sadness, Guilt, Hostility (Anger), Shyness, Fatigue, Surprise, Joviality, Self-Assurance, Attentiveness, and Serenity, it is not able to measure discrete emotions such as PLAY. To obtain objective data, EEG and ECG machines are used in order to see which parts of the brain are stimulated.

In Study 1, done by Ming Li et al., the participants involved twenty eight healthy volunteers aged twenty one to thirty. Each participant was instructed to sit quietly in a chair for five minutes to obtain the baseline data. The subject's 9-point SAM score was recorded. The subjects were then showed eight different virtual reality videos at random. After each video which was about three minutes and thirty seconds long, they were asked to rate their emotions and sense of presence using the SAM and IPQ questionnaires which were displayed on the screen to prevent the disturbance of the EEG electrodes [13]. It would take the subjects about five minutes to fill in the questionnaire which would also serve as a break to prevent mental fatigue. After collecting the data, EEGLAB,

an open-source Matlab toolbox that provides powerful algorithms for EEG processing, feature extraction, and emotion recognition was used to obtain clean and high quality EEG signals for frequency domain feature extraction [13].

In Study 2 done by Susindar et al, there were eleven participants between the age of 19 and 29. The first part of the study involved obtaining baseline data by asking the participants how they have been feeling “in the past hours”. This is when they filled their first set of PANAS-X responses. They were then asked to practice guided breathing while listening to nature sounds for five minutes. A second set of PANAS-X responses were collected for participant’s “current feelings”, this is when the baseline physiological data was collected. The participants were then exposed to stimulus lasting about 5 and a half minutes and asked to record their responses on another PANAS-X questionnaire. Following this, they were asked to perform the decision-making task on the PEBL-BART platform. This part was repeated for the other remaining emotional stimuli on the other display type. Each participant was presented a combination of presentation medium and target emotion in two parts. Once this was completed, they were asked to practice guided breathing while listening to nature sounds for two minutes [25].

4.5 Challenges to emotion elicitation in VR: Cybersickness

Despite the efficacy of Virtual Reality (VR) as an emotional medium, its potential to induce cybersickness remains a primary barrier to successful elicitation. If a user experiences physical discomfort, the intended transition into a state of social joy or PLAY may be fundamentally obstructed. Cybersickness typically arises from sensory conflict, where the brain detects a mismatch between visual, vestibular (balance), somatosensory, and proprioceptive cues. To mitigate this, researchers often attempt to "re-weight" these inputs, specifically by down-weighting vestibular signals in favor of visual information [26].

Beyond basic sensory integration, the severity of cybersickness is influenced by several intersecting factors:

Content and Visual Characteristics. Features such as optic flow, visual lag, and the field of view (FOV) significantly impact the user experience. While high-fidelity environments are desirable, the absence of sickness-mitigation features, such as rest frames, can exacerbate the feeling of vection (the illusion of self-motion) and lead to rapid-onset discomfort [26].

Interaction and Navigation. The way a user moves through the Virtual Environment (VE) is a critical determinant of sickness. Task-related features including rotational velocity, sudden acceleration, and the overall controllability of the navigation style must be carefully calibrated to prevent overwhelming the user’s equilibrium[26].

Human Factors. Susceptibility is highly individualized and can be categorized into intrinsic factors, such as gender and inherent motion-sickness sensitivity, and extrinsic factors, such as the user’s level of prior video game experience[26].

Hardware Constraints. While it is often assumed that technological advancements in Head-Mounted Displays (HMDs) would naturally eliminate these issues, research suggests that cybersickness does not necessarily decrease linearly with improved hardware[26].

The most common method for measuring cybersickness is the use of subjective questionnaires, specifically the Simulator Sickness Questionnaire (SSQ). This tool assesses cybersickness across three specific symptom clusters: Nausea, Oculomotor strain, and Disorientation, utilizing a 4-point Likert scale. While other instruments have been developed to evaluate motion sickness such as the Motion Sickness Assessment Questionnaire (MSAQ), it remains unclear whether these are sufficiently accurate for specifically measuring cybersickness in virtual environments. The SSQ has previously been criticised due to its psychometric properties. Variants of the SSQ such as the Virtual Reality sickness questionnaire(VRSQ) have been developed with the aim of re-framing the questions from the SSQ[26].

The cybersickness susceptibility questionnaire(CSSQ) was proposed to collect demographic data, health, fitness, and motion sickness susceptibility[10]. A strong correlation was found between CSSQ items and cybersickness.

In addition to subjective data obtained from questionnaires, objective data can be collected through various physiological measurements. Past research has attempted to link physiological markers such as the Electrocardiogram (ECG), Electroencephalogram (EEG), Electrodermal Activity (EDA), and Electrogastrography (EGG) to the user experience. Using questionnaires in conjunction with these physiological measurements ensures that biological changes are directly related to what the user was experiencing. Furthermore, the collection of physiological data allows researchers to observe these changes continuously without interrupting the user’s immersion or sense of presence[26].

Cybersickness can be prevented or at the very least reduced by narrowing the field of view in the vertical axis or decreasing the field of view to sixty degrees. Another effective method is the combination of a rest frame display with a dynamic reduction of the field of view. Additionally, a blur effect can be used where non salient objects in the scene are detected and blurred in the display image. Beyond visual adjustments, postural susceptibility during VR gameplay also serves as a reliable indicator for individual susceptibility[26].

Although VR is an effective medium for emotion elicitation, measures must be implemented to ensure that participants remain comfortable and that the targeted emotion can be successfully elicited.

4.6 Previous Work

This section discusses two different papers that were done to evaluate the effectiveness of VR as an emotion elicitation medium.

The goal of the experiment. The experiment done by Ming Li et al. aimed to elicit emotions in the four quadrants of Russell’s arousal-valence model [22]. Whereas the experiment done by Susindar et al, was done to assess the influence of target emotions elicited by VR on decision making behavior [25]. Both experiments were done to investigate the effects of VR on emotion elicitation.

Participants. Ming Li et al. had 28 volunteers between the ages of 21 to 33 with 16 males and 12 females. Susindar et al. had 11 volunteers between the ages of 19 and 29 with 5 females and 6 males. Both experiments had a limited age distribution suggesting that their findings may not apply to the wider population group.

Measurements taken. Ming Li et al. collected subjective data from participants using the 9-point SAM scale and the IPQ score to measure the level of presence the participants experience during the experience while Susindar et al. used the PANAS-X scale and a sickness questionnaire. To obtain objective data, Ming Li et al. measured the EEG and ECG signals of participants during the experiment while Susindar et al. measured Skin Conductance Response (SCR) and Heart Rate (HR). The collection of EEG data revealed the power of β and γ when dealing with high-valence emotions. The participants in Susindar et al.'s experiment also performed The Balloon Analog Risk Task (BART) which revealed the impact the emotions elicited had on decision making. Although the use of EEG to collect physiological data provided more data, due to the fact that EEG devices are sensitive to movement, the participants movements were discouraged and as a result may have limited their experience.

Time to recover between stimuli. The participants in Ming Li et al.'s experiment were given a five minute break between each of the eight videos which was about three minutes and thirty seconds long. On the other hand, participants in Susindar et al.'s experiment were given a two minute break between the two videos where each was about 5 and a half minutes long. The break in the study done by Ming Li et al. may not have been long enough as participants were required to watch eight videos, this may have led to mental fatigue.

Obtaining baseline data. The participants in the experiment done by Ming Li et al. were asked to sit in a chair for five minutes in order to obtain baseline data while the participants in the experiment done by Susindar et al. were required to breathe for five minutes.

Both studies asked the participants about their previous experiences with VR. This was important as Ming Li et al. found that participants who used VR for the first time were excited and as a result had higher levels of arousal.

5 Conclusion

This literature review set out to examine whether Virtual Reality (VR) is a viable medium for eliciting Panksepp's primary emotional system of PLAY. To this end, three bodies of literature were reviewed: the neurobiological basis and clinical importance of PLAY, the principles of emotion elicitation, and the effectiveness of VR as an emotion elicitation medium.

PLAY is a well-established emotion as described by Jaak Panksepp, the subcortical circuits within the periaqueductal gray area (PAG) and the lower limbic system are responsible for the emotion. Therapeutic approaches that promote the positive hedonics of PLAY provide alternative treatment options due to the underlying neuroanatomy of PLAY.

The literature reveals VR as an effective medium in emotion elicitation. This is due to its ability to create a high sense of presence which is directly related to the emotional experience of the user. VR engages the physiological and psychological components of the user making the experience more immersive. Furthermore, established design principles combining semantic, dynamic, and interactive elements provide a framework for constructing VR environments capable of targeting specific emotional states.

Measuring PLAY elicitation requires a multimodal approach combining the ANPS scale specifically used to measure Panksepp's primary emotional systems with physiological measures such as EEG and ECG. However, since the ANPS was designed to measure stable emotional traits rather than momentary states, it should be supplemented with state-level measures such as the PANAS-X to capture the immediate emotional response to VR stimuli.

Despite the growing body of literature on VR-based emotion elicitation and the well-established neurobiological basis of PLAY, no study to date has specifically designed a VR environment to target Panksepp's PLAY system as a discrete primary emotion. This gap is clinically significant given PLAY's documented links to depression through the GRIEF and SEEKING systems. The present research addresses this gap by proposing three VR environments designed to elicit PLAY, evaluated through a multimodal measurement approach, representing a theoretically motivated and clinically relevant contribution to both affective neuroscience and VR research.

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